

TTS Reports V1.0 MVP Assessment

MVP suitability

Suitable for MVP testing: Yes

The current build is suitable for:

- internal demonstration
- controlled user testing
- staff testing
- testing with real Metrel project files
- comparing against PATLink PC / ES Manager reporting
- validating whether customers understand the workflow
- checking whether the generated reports are useful enough

Not yet suitable for full public SaaS launch

It is not yet a full commercial cloud product because it still runs locally and does not include:

- user accounts
- cloud storage
- hosted deployment
- billing
- admin dashboard
- permissions
- long-term file/report history
- customer database
- automated updates

But that is fine. Those are **future expansion features**, not MVP requirements.

Current Features Implemented

1. Local Web App

The app runs locally as a Flask web app.

Users can start it with:

```
python app.py
```

Then access it through:

http://127.0.0.1:5000

This is suitable for V1.0 because it avoids cloud hosting, user accounts, security overhead, and deployment complexity.

2. Multi-format Project Import

The app supports importing project data from several Metrel-related file types.

Supported import types:

.pdf
.padfx
.apx
.xlsx

Current file source explanations:

File type	Source
.pdf	Project report exported directly from the Metrel aPAT Android Application
.padfx	Project export from the Metrel aPAT Android Application
.apx	Project export / MESM file from the Metrel aPAT Android Application
.padfx	Saved project file from a project imported into the Metrel ES Manager Software
.xlsx	Exported project spreadsheet from the Metrel ES Manager Software

This is one of the strongest MVP features because it gives users flexibility depending on what export format they already have.

3. Asset Register Dashboard

The Home page now works as a clean **Asset Register Dashboard**.

Implemented features:

- imported asset table
- asset number display
- location display
- description/item display

- status/result display
- retest/date information where available
- tested by/operator column
- dashboard search
- status filter
- location filter
- summary cards/statistics
- selected asset modal instead of a fixed side panel

This makes the Home page feel like a proper web app rather than just an upload form.

4. Asset Details Modal

When a user selects an asset, the app now opens the asset details in a popup modal.

Implemented modal features:

- click asset row to open details
- close button
- click outside modal to close
- press **Esc** to close
- keeps the dashboard clean
- avoids the old Selected Asset panel taking up space

This is a good UX improvement for the MVP.

5. Table View and Tree View

The Asset Register Dashboard now supports two view modes:

Table View

Tree View

Table View

Standard asset register table.

Useful for:

- scanning all assets
- filtering
- reviewing imported data quickly

Tree View

Assets are grouped directly by **Location**.

Example:

Kitchen

- Microwave
- Kettle
- Toaster

Office

- Computer
- Monitor
- Power board

Implemented Tree View features:

- group assets by location
- expand/collapse location groups
- Expand All button
- Collapse All button
- click asset to open asset detail modal

This is very useful for test and tag users because location-based asset grouping matches how many jobs are physically performed.

6. Reports Page Only Workflow

Report generation controls were removed from the Home page.

Current workflow:

Import file → Review Asset Register → Go to Reports → Preview/Generate report

This is cleaner and more logical.

Implemented:

- Home page is now dashboard-focused
- Reports page contains report controls
- report generation only happens from Reports
- fewer accidental submissions
- better separation between reviewing data and producing reports

This is MVP-appropriate.

7. Report Types

The app currently supports multiple report types.

Basic Report

A simpler customer-facing report.

Implemented features:

- richer title page
- customer/site details
- contractor details
- generated date
- test standard
- report basis
- testing start/end information
- next inspection date
- operator/tester field
- asset count summary
- passed/failed/other summary
- result statement
- notes section
- client/operator sign-off area
- simple report body

This is suitable for customers who want a clear, less detailed report.

PRO Report

A more detailed report option.

Implemented features:

- professional title page
- detailed test result information
- asset/test result tables
- customer and contractor details
- more complete report layout

This is likely the main paid/value report type.

Upcoming Retest Report

A forward-looking report showing assets due for retesting.

Useful for:

- follow-up work
- customer reminders
- recurring revenue opportunities

- compliance planning

Advanced / Audit Report

The app supports an expanded Advanced/Audit reporting mode.

Implemented features include:

- advanced controls on Reports page
- scrollable controls panel
- access to extra report configuration
- date range style workflow
- historical/all-data style reporting concept

This is useful for internal/audit-style use, but should probably remain secondary to Basic and PRO.

8. Latest Test Only Logic

The app includes a **Latest Test Only** concept.

This is very important.

Why it matters:

- imported project files may contain multiple historical tests for the same asset
- customer reports should normally show the most recent test only
- older historical records can make reports confusing
- it avoids duplicate-looking assets/results

For MVP, this is a key feature.

9. Report Preview

The app now includes a report preview workflow.

Implemented:

- Preview Report button
- preview opens inside the app
- modal-style preview window
- close preview button
- click outside / Esc close behaviour
- Export PDF from preview
- Export Preview PDF button
- direct Generate PDF Report button still available

This is a strong MVP feature because users can check the report before exporting.

10. PDF Report Export

Implemented:

- generate PDF reports
- export/download PDF
- reports saved/generated through the app
- output folder support
- browser download support

This is the core deliverable of the product.

11. Simplified Settings Page

The Settings page has been cleaned up and reduced to V1.0-relevant options.

Current tabs:

Business

Report Defaults

Report Template Defaults

Sign-off

Removed/hid from V1.0:

- Branding
- Retest settings
- Import settings
- PDF settings
- Privacy settings
- Advanced/developer settings

This makes the MVP cleaner and less confusing.

12. Business Settings

Implemented business profile settings:

- business name
- trading name
- ABN / ACN
- electrical licence number
- phone

- email
- website
- address
- default technician/operator

These support contractor details on reports.

13. Report Defaults

Implemented simplified report defaults.

Likely includes:

- default report type
- default standard, for example [AS/NZS 3760](#)
- include detailed appliance pages

These are appropriate for V1.0.

14. Report Template Defaults

Implemented template defaults for report prefill.

Includes fields such as:

- customer number
- inspection record prefix
- order number prefix
- description
- reason for test
- equipment type
- attachments
- labelling method
- notes

This supports faster report preparation.

15. Sign-off Settings

Implemented sign-off related settings:

- client location
- client date
- client signature/name

- operator location
- operator date
- operator signature/name

This supports the title page / declaration section.

16. Help Page

The app now includes a Help page under Settings.

Implemented Help content:

- quick setup guide
- how to start the app
- how to import files
- where each file type comes from
- how to use the Asset Register Dashboard
- how to use the asset modal
- how to complete Template and Settings details
- how to generate reports
- Basic / PRO / Upcoming Retest report explanations
- Latest Test Only vs Advanced/Audit explanation
- where reports are saved
- common troubleshooting
- recommended V1 workflow

This is very useful for MVP testing because users can self-orient without needing support immediately.

17. Windows Build Support

The app includes a Windows executable build process using:

`build_windows.bat`

The build process was also improved by recommending use of a clean virtual environment to avoid PyInstaller scanning unrelated packages.

This supports eventual simple local distribution.

Current MVP Workflow

The current intended workflow is:

1. Open TTS Reports
2. Import Metrel/aPAT/ES Manager project file
3. Review imported assets on the Asset Register Dashboard
4. Search/filter assets if needed
5. Use Table View or Tree View
6. Click assets to inspect details in modal
7. Go to Template/Settings if needed
8. Go to Reports
9. Choose Basic, PRO, Retest, or Advanced/Audit
10. Preview the report
11. Export PDF
12. Send PDF to customer

That is a complete MVP workflow.

Features That Could Be Added in Future Expansion

Near-term V1.1 Improvements

These are practical improvements after MVP testing.

1. Better Report Preview

Current preview is useful, but future improvements could include:

- full page-by-page PDF preview
- zoom controls
- print preview
- report page thumbnails
- warnings before export
- missing-field alerts

Example:

Warning: No customer email entered.

Warning: 3 assets have no retest date.

Warning: 2 failed assets found.

2. Report Validation Checklist

Before exporting, the app could check:

- customer name entered
- site/location entered
- contractor details completed
- operator/tester present
- test standard selected
- asset count looks correct
- failed assets flagged
- missing retest dates
- imported file format identified

This would reduce support issues.

3. Better File Import Summary

After import, show a summary such as:

File imported successfully

Format: aPAT PDF Export

Total assets: 125

Passed: 119

Failed: 4

Other: 2

Locations found: 8

Testers found: 2

Date range: 01/05/2026 - 05/05/2026

This would help users trust the import.

4. Import Error Handling

Add clearer messages for:

- unsupported files
- blank files
- PDF with no selectable text
- missing columns in spreadsheet
- old export formats
- duplicate records
- no assets found
- no latest test identified

This will be very important as more real-world files are tested.

5. Editable Asset Data Before Report

Allow users to correct imported data before generating the report.

Examples:

- edit location
- edit asset description
- edit asset ID
- edit customer/site details
- edit tester name
- correct retest date
- mark asset as excluded from report

This would make the app much more flexible.

6. Exclude Assets From Report

Add checkboxes to exclude selected assets.

Useful when:

- imported project contains assets not required for this customer report
 - old/irrelevant assets appear
 - user wants to hide abandoned/deleted equipment
 - failed/removed assets need separate handling
-

7. Save Imported Project Session

Allow the user to save the current imported data/session.

For local app:

Save Project
Open Recent Project
Clear Project

For cloud version:

Customer > Site > Project > Reports

8. Improved Retest Report

Future Retest Report features:

- group by overdue / 30 / 60 / 90 days
- group by customer
- group by site
- group by location
- export retest CSV
- email reminder template
- retest quote generator
- retest booking checklist

This could become a very useful sales/support feature.

9. Failed Asset Report

Add a dedicated report for failed assets only.

Could include:

- failed asset list
- failure reason
- location
- risk/action required
- repair/retest recommendation
- customer sign-off section

This is commercially useful because failed assets often require follow-up action.

10. Certificate / Compliance Summary Page

Add an optional certificate-style page at the front.

Could include:

Electrical Safety Testing Summary
Test Standard: AS/NZS 3760
Total assets tested: 124
Passed: 121
Failed: 3
Testing completed by: [Operator]
Date completed: [Date]

This could make the Basic Report feel more official.

Medium-term V1.5 Features

11. Customer Database

Add local customer records:

- customer name
- contact person
- email
- phone
- site address
- notes
- previous reports
- retest schedule

This would reduce repeated data entry.

12. Site / Location Management

A customer could have multiple sites.

Example:

Customer: ABC Pty Ltd

Site: Brisbane Office

Site: Gold Coast Warehouse

Site: Toowoomba Depot

This becomes important once report history and cloud storage are added.

13. Report History

Store generated reports in a searchable history.

Fields:

- customer
- site
- date generated
- report type
- number of assets
- result summary
- source file
- generated by

This would make the tool feel like proper reporting software.

14. Branded Report Themes

Add controlled branding options:

- logo upload
- primary colour
- secondary colour
- cover page style
- footer text
- company slogan
- report watermark
- TTS default theme
- contractor custom theme

This should come after report layout is stable.

15. Multiple Report Templates

Add report template choices:

Basic Customer Report
Detailed PRO Report
Failed Assets Report
Upcoming Retest Report
Audit Report
Certificate Summary
Internal Asset Register

Users could choose what suits the customer.

16. CSV Export

Allow users to export cleaned/imported data as:

.csv
.xlsx

Useful for:

- internal records
 - customer asset registers
 - audits
 - importing into other systems
-

17. Bulk Report Generation

Allow users to generate multiple reports from one import.

Example:

Generate:

- Customer Basic Report
- Detailed PRO Report
- Failed Assets Report
- Upcoming Retest Report

All exported at once.

18. Email-ready Outputs

Generate email copy with the report.

Example:

Hi [Customer],

Please find attached your electrical test and tag report for [Site].

Summary:

- Total assets tested: 124
- Passed: 121
- Failed: 3
- Next inspection due: [Date]

Regards,
[Business Name]

This would save users time.

19. Quote / Invoice Integration

Possible future add-ons:

- quote for failed asset replacements
- quote for upcoming retest
- consumables used
- labels/tags used
- invoice summary
- Xero/MYOB export

This could turn TTS Reports into a business workflow tool, not just a report generator.

Long-term V2.0 Expansion

20. Hosted Web App

Move from local Flask app to hosted web app.

Features:

- customer login
- contractor login
- secure file upload
- cloud storage
- access from tablet, PC, or phone
- no local installation
- always updated

This aligns strongly with your original idea: customers exporting from aPAT on a tablet, opening TTS Reports in a browser, and generating the customer report without a PC.

21. User Accounts

Add accounts for:

- individual contractors
- businesses
- staff/team members
- admin users
- customer read-only users

This supports SaaS pricing.

22. Cloud Storage

Store:

- imported project files
- parsed asset data
- generated reports
- customers
- sites
- templates
- branding
- report history

This becomes the foundation for a full reporting platform.

23. Subscription Plans

Possible plans:

Free Trial
Basic
Pro
Business
Enterprise

Example structure:

Plan	Features
Basic	Basic reports, limited imports
Pro	PRO reports, retest reports, branding
Business	multiple users, report history, customer database
Enterprise	team management, integrations, bulk reporting

24. Customer Portal

Let end customers access reports.

Possible features:

- view past reports
- download certificates
- see upcoming retest dates
- approve quotes
- request retesting
- view failed assets

This could create long-term lock-in.

25. Retest Reminder System

Automated reminders:

- contractor reminders
- customer reminders

- due soon alerts
- overdue alerts
- monthly retest schedule
- email notifications

This is a strong commercial expansion because it helps contractors generate repeat work.

26. Product/Consumables Marketing Section

Add a small non-intrusive marketing area.

Could promote:

- test tags
- printers
- scanners
- adaptors
- replacement leads
- Metrel accessories
- specials
- service updates
- software tips

This fits your earlier internal business use case.

27. TTS Admin Portal

For your business internally:

- manage announcements
 - manage promoted products
 - view customer usage
 - see common import errors
 - see most used report types
 - publish app updates
 - manage help content
-

28. AI-assisted Report Review

AI could check imported project/report before export.

Examples:

This report contains 4 failed assets.
12 assets are missing retest dates.
The customer name is blank.

The date range includes old historical tests.
There are duplicate asset IDs.

This would be helpful without generating the report content for the user.

29. AI Help Assistant

A built-in assistant trained on:

- how to export from aPAT
- how to export from ES Manager
- supported file types
- common import errors
- how to generate reports
- report type explanations

This could reduce support calls.

30. Integration With Metrel Workflow Guides

Add step-by-step export guides:

How to export PDF from aPAT

How to export .padfx from aPAT

How to export .apx MESM file from aPAT

How to export .xlsx from ES Manager

How to save .padfx from ES Manager

Could include screenshots or short videos later.

Recommended MVP Testing Plan

I would test this build with a small group before adding more features.

Test group

Start with:

3 to 5 internal users

3 to 5 trusted customers

Test files

Use real examples:

- aPAT PDF export
- aPAT `.padfx`
- aPAT `.apx`
- ES Manager `.padfx`
- ES Manager `.xlsx`

Ask testers

Key questions:

1. Could you import your project file without help?
 2. Did the Asset Register look correct?
 3. Did the Tree View help?
 4. Was the report preview useful?
 5. Was the Basic Report suitable for customers?
 6. Was the PRO Report better than the existing software output?
 7. Did you understand Latest Test Only?
 8. Did you know where to go to generate reports?
 9. Was the Help page enough?
 10. Would this save time compared with PATLink PC / ES Manager?
 11. Would you use it if it was web-based?
 12. Would you pay for it or expect it as a TTS support tool?
-

My Recommendation

Do **not** add many more features before MVP testing.

The current build is already enough to test the core product.

The only features I would consider before user testing are:

1. clearer import success summary
2. report validation warnings before export
3. improved error messages for bad imports
4. one or two sample files included for demo/testing
5. a small "V1.0 MVP" label/version number in the footer or Help page

Everything else should wait until after real users test it.

Your current MVP is now focused around the right promise:

Import Metrel project data, review the asset register, preview the report, and export a cleaner customer-ready PDF without fighting the existing reporting software.

